

Memory Hierarchy

COMS10015

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Week 17

Memory requirements: programs

In a von Neumann architecture, programs (and data) are stored in memory.

- ▶ Processor accesses memory in order to know what operation to do.
- ▶ Want this to be quick so processor not waiting for work.
- ▶ Want it to be large enough to hold the programs.
- ▶ Want it to be cheap to make.

Memory requirements: data

Also need memory to store data.

- ▶ Want large capacity as need a lot of data.
- ▶ Want it to be quick: low latency and high bandwidth.
- ▶ Want it to be cheap to make.

Processor has relatively few registers, so need bigger memory for more storage.

- ▶ Can give memory addresses directly to instructions — recall lecture on ISA addressing modes.

Memory technologies

We can build memory out of different technologies:

- ▶ Flip-flops.
- ▶ SRAM.
- ▶ DRAM.
- ▶ High-bandwidth DRAM (HBM, MCDRAM): 3D stacked DRAM.
- ▶ Flash.
- ▶ Hard disk.
- ▶ Tape.

Each has different characteristics.

SRAM vs DRAM

SRAM

- ▶ Volatile.
- ▶ Faster access (no capacitor).
- ▶ Lower density (6 transistors per cell).
- ▶ Higher cost.
- ▶ No need for refresh.
- ▶ Easy to manufacture along with logic gates.

DRAM

- ▶ Volatile.
- ▶ Slower access (capacitor).
- ▶ Higher density (1 transistor and 1 capacitor per cell).
- ▶ Lower cost.
- ▶ Requires refresh.
- ▶ Manufacturing involved both logic (transistors) and capacitors.

Memory costs

Faster is more expensive

- ▶ SRAM, <\$10 per Megabyte.
- ▶ DRAM, <\$1 per Megabyte.
- ▶ Hard Disk, <\$1 per Gigabyte.

Prices circa 2011, from Onur Mutlu Lecture 23b.

Bigger is slower

- ▶ SRAM, Kilobytes–megabytes (10^3 – 10^6), up to tens of nanoseconds (1–50 cycles).
- ▶ DRAM, Gigabytes (10^9), around 100 nanoseconds (~ 200 cycles).
- ▶ Hard Disk, Terabytes (10^{12}), tens of milliseconds.

Cycles to nanoseconds

A 2.2 GHz processor has 2.2×10^9 cycles per second; each cycles takes 0.45 ns.

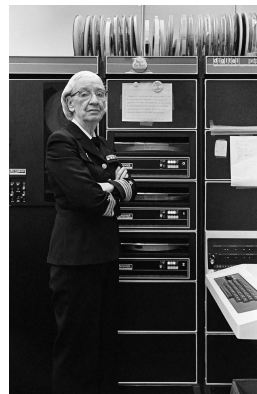
Grace Hopper

How long is a nanosecond?

11.8 inches

Derived from how far light can travel in a wire in 1 nanosecond.

- ▶ Dr Grace Hopper (1906–1992) was American Computer Scientist.
- ▶ PhD in mathematics from Yale, Professor at Vassar College, joined US Navy.
- ▶ Invented one of the first linkers, converting English-like names of subroutines into machine code.
- ▶ Developed early imperative programming languages.



Picture from Wikipedia CC BY-SA 4.0 Grace Murray Hopper, in her office in Washington DC, 1978, ©Lynn Gilbert

Memory requirements

Unfortunately, have to make a trade-off:

- ▶ Bigger is slower — takes longer to find location.
- ▶ Faster is more expensive — due to memory technology.
- ▶ High bandwidth is more expensive — more wires.

Caches

With main memory so far away (~ 200 cycles), instructions and data would take a long time to fetch.

- ▶ Recall: pipelined processor fetches an instruction *every cycle*.
- ▶ Need a closer memory to keep upcoming instructions in.
- ▶ Processors designed with small instruction **caches**.

Definition: Cache

A cache is a memory that stores data so that future requests to that data is faster.

Instruction caches

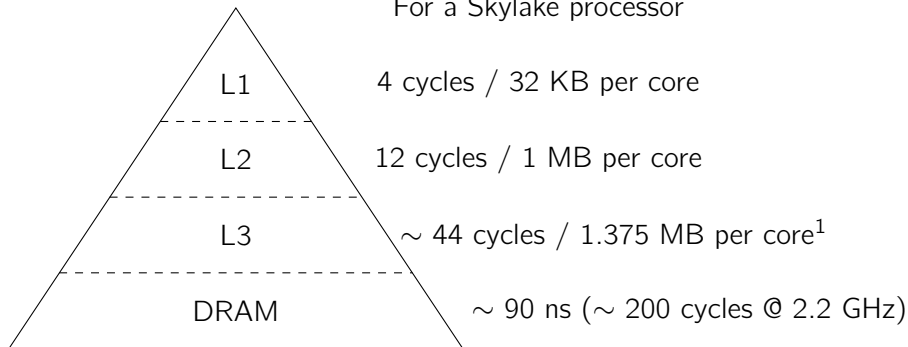
Most modern processors have a dedicated instruction cache.

- ▶ Typical capacity is 32 KB.
- ▶ If instructions are 64-bits, can save 4,096 instructions.
- ▶ Real programs will be significantly larger, so cache only stores a portion.
- ▶ Example: a loop will repeat the same few instructions many times. If a copy of those instructions are saved in the instruction cache, there is no need to wait for them to be repeatedly fetched from main memory.
- ▶ Remainder of program stored in main memory, copied in chunks to the cache as needed.
- ▶ The micro-architecture has mechanisms to do all of this.

Data caches

Processors also implement data caches to help improve data access times.

For a Skylake processor



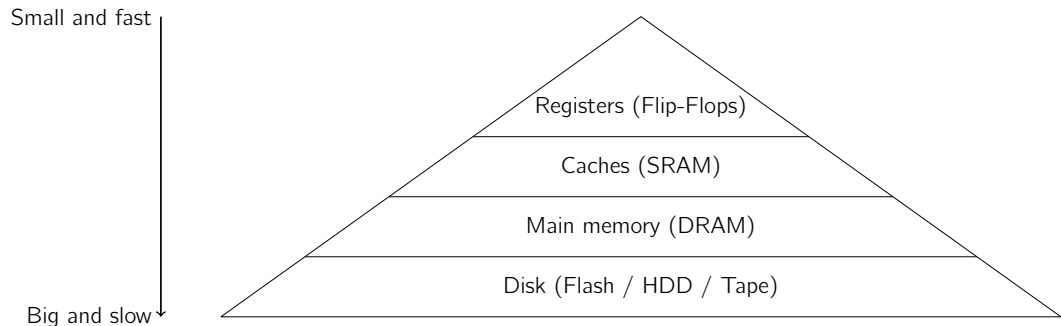
¹L3 is a large resource shared between all cores. In contrast, each core has own private L1 and L2.

Data caches

Data caches work similarly to the instruction cache.

- ▶ Commonly used data is held close to the processor.
- ▶ Usually separate from instruction cache so they don't interfere with each other.
- ▶ Commonly two or three levels of data caches.
- ▶ For consistency, we say L1 is the lowest (or closest) level of cache.

Memory hierarchy



Memory hierarchy in practice

We still use all these memories at the same time, even tape (which were first used in 1950s).

Example: Trinity Supercomputer, at Los Alamos National Laboratory, US:

- ▶ 19,420 nodes (9,436 Haswell + 9,984 KNL).
- ▶ Each Haswell node has 2 CPUs (which has registers and caches) and 128 GB DRAM.
- ▶ Each KNL node has 1 CPU (registers and caches), 96 GB DRAM and 16 GB of MCDRAM (3D stacked DRAM).
- ▶ That's 2,326,016 GB or 2.3 PB (1 PB is 10^{15} bytes) of *working memory*.
- ▶ Plus 3.7 PB flash memory (for quick, semi-temporary storage).
- ▶ Plus 78 PB filesystem of hard disks.
- ▶ Plus > 100 PB of long term tape storage.

More info: <https://www.osti.gov/servlets/purl/1457905> and [https://en.wikipedia.org/wiki/Trinity_\(supercomputer\)](https://en.wikipedia.org/wiki/Trinity_(supercomputer))

Further Reading

- ▶ Grace Hopper: https://en.wikipedia.org/wiki/Grace_Hopper
- ▶ Grace Hopper lecture: <https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3884041/nsa-releases-copy-of-internal-lecture-delivered-by-computing-giant-rear-adm-gra/>
- ▶ High bandwidth memory: https://en.wikipedia.org/wiki/High_Bandwidth_Memory
- ▶ L23b, L24 and L25a: <https://safari.ethz.ch/digitaltechnik/spring2018/doku.php?id=schedule>
- ▶ L15 and L16: Caches <https://courses.cs.washington.edu/courses/cse378/09wi/lectures/lec15.pdf> and <https://courses.cs.washington.edu/courses/cse378/07au/lectures/L16-More-Cache-Organizations.pdf>
- ▶ CPU Cache: https://en.wikipedia.org/wiki/CPU_cache